Supplemental Material: Hierarchical structuring of weapon allometry in the European earwig.

Céline Ozkan[2] and Clint D. Kelly[1,✉]

[1] Département des Sciences biologiques, Université du Québec à Montréal, Montréal, Canada

[2] Université Claude Bernard Lyon 1, Lyon, France

✉ Correspondence: Clint D. Kelly <kelly.clint@uqam.ca>

Statistical models

Sex difference model

We modelled log forceps length as a linear function of body size, sex, diet, and density, including all lower-order terms and interactions among predictors. Model structure:

$$\log F_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$$

$$\begin{aligned}\mu_{ij} = {} & \beta_0 + \beta_1 \log P_{ij} + \beta_2 \text{sex}_{ij} + \beta_3 \text{diet}_{ij} + \beta_4 \text{density}_{ij} + \beta_5(\log P_{ij} \times \text{sex}_{ij}) \\ & + \beta_6(\log P_{ij} \times \text{diet}_{ij}) + \beta_7(\log P_{ij} \times \text{density}_{ij}) + \beta_8(\text{sex}_{ij} \times \text{diet}_{ij}) \\ & + \beta_9(\text{sex}_{ij} \times \text{density}_{ij}) + \beta_{10}(\text{diet}_{ij} \times \text{density}_{ij}) \\ & + \beta_{11}(\log P_{ij} \times \text{sex}_{ij} \times \text{diet}_{ij}) + \beta_{12}(\log P_{ij} \times \text{sex}_{ij} \times \text{density}_{ij}) \\ & + \beta_{13}(\log P_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}) + \beta_{14}(\text{sex}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}) \\ & + \beta_{15}(\log P_{ij} \times \text{sex}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}) + b_{0j} + b_{1j} \log P_{ij}\end{aligned}$$

where $\varepsilon_{ij} \sim \text{Normal}(0, \sigma^2)$, and maternal identity (j) has a varying intercept (b_{0j}) and slope (b_{1j}). Maternal identity was modelled as a random effect with both intercept and slope variation with respect to body size, allowing among-family differences in both mean trait size and allometric scaling. Each coefficient β corresponds to the matching main effect or interaction term reported in Tables S1–S2 (e.g., β_5 corresponds to logP × sex, β_{11} to logP × sex × diet).

Test for non-linearity

To test for curvature in allometry, we added a quadratic term for body size: Model structure:

$$\log F_{ij} \sim \text{Normal}(\mu_{ij}, \sigma)$$

$$\mu_{ij} = \beta_0 + \beta_1 \log P_{ij} + \beta_2 \left(\log P_{ij}\right)^2 + \beta_3 \text{sex}_{ij} + \beta_4 \text{diet}_{ij} + \beta_5 \text{density}_{ij} + \beta_6 \left(\log P_{ij} \times \text{sex}_{ij}\right)$$
$$+ \beta_7 \left(\log P_{ij} \times \text{diet}_{ij}\right) + \beta_8 \left(\log P_{ij} \times \text{density}_{ij}\right) + \beta_9 \left(\left(\log P_{ij}\right)^2 \times \text{sex}_{ij}\right)$$
$$+ \beta_{10} \left(\left(\log P_{ij}\right)^2 \times \text{diet}_{ij}\right) + \beta_{11} \left(\left(\log P_{ij}\right)^2 \times \text{density}_{ij}\right) + \beta_{12} \left(\text{sex}_{ij} \times \text{diet}_{ij}\right)$$
$$+ \beta_{13} \left(\text{sex}_{ij} \times \text{density}_{ij}\right) + \beta_{14} \left(\text{diet}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{15} \left(\log P_{ij} \times \text{sex}_{ij} \times \text{diet}_{ij}\right) + \beta_{16} \left(\log P_{ij} \times \text{sex}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{17} \left(\log P_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right) + \beta_{18} \left(\left(\log P_{ij}\right)^2 \times \text{sex}_{ij} \times \text{diet}_{ij}\right)$$
$$+ \beta_{19} \left(\left(\log P_{ij}\right)^2 \times \text{sex}_{ij} \times \text{density}_{ij}\right) + \beta_{20} \left(\left(\log P_{ij}\right)^2 \times \text{diet}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{21} \left(\text{sex}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right) + \beta_{22} \left(\log P_{ij} \times \text{sex}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{23} \left(\left(\log P_{ij}\right)^2 \times \text{sex}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right) + b_{0j} + b_{1j} \log P_{ij}$$

This model tests whether scaling relationships deviate from linearity.

Male morph model

We modelled male forceps length as a function of body size, morph, diet, and density: Model structure:

$$\log F_{ij} \sim \text{Normal}\left(\mu_{ij}, \sigma\right)$$

$$\mu_{ij} = \beta_0 + \beta_1 \log P_{ij} + \beta_2 \text{morph}_{ij} + \beta_3 \text{diet}_{ij} + \beta_4 \text{density}_{ij} + \beta_5 \left(\log P_{ij} \times \text{morph}_{ij}\right)$$
$$+ \beta_6 \left(\log P_{ij} \times \text{diet}_{ij}\right) + \beta_7 \left(\log P_{ij} \times \text{density}_{ij}\right) + \beta_8 \left(\text{morph}_{ij} \times \text{diet}_{ij}\right)$$
$$+ \beta_9 \left(\text{morph}_{ij} \times \text{density}_{ij}\right) + \beta_{10} \left(\text{diet}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{11} \left(\log P_{ij} \times \text{morph}_{ij} \times \text{diet}_{ij}\right) + \beta_{12} \left(\log P_{ij} \times \text{morph}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{13} \left(\log P_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right) + \beta_{14} \left(\text{morph}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right)$$
$$+ \beta_{15} \left(\log P_{ij} \times \text{morph}_{ij} \times \text{diet}_{ij} \times \text{density}_{ij}\right) + b_{0j} + b_{1j} \log P_{ij}$$

Table S1. Posterior summaries from the Bayesian sex-specific allometry model. The model estimated Gaussian regression of log forceps length including body size (logPc), sex, diet, density, and their interactions, with varying intercepts and slopes for mother identity (id_mere). Reported values are posterior medians and 95% credible intervals (CrI).

Section	Parameter	Estimate	95% CrI
Fixed effects	Intercept	1.07	0.9, 1.25
Fixed effects	logP	0.35	0.06, 0.64
Fixed effects	Sex (male)	-0.06	-0.28, 0.15
Fixed effects	Diet (poor)	-0.02	-0.25, 0.22
Fixed effects	Density = 4	0.03	-0.19, 0.25
Fixed effects	Density = 8	0.07	-0.13, 0.28
Fixed effects	logP × Sex (male)	0.61	0.25, 0.98
Fixed effects	logP × Diet (poor)	-0.07	-0.52, 0.37
Fixed effects	Sex (male) × Diet (poor)	0.29	-0.01, 0.58
Fixed effects	logP × Density = 4	-0.02	-0.39, 0.35
Fixed effects	logP × Density = 8	-0.09	-0.42, 0.25
Fixed effects	Sex (male) × Density = 4	-0.16	-0.45, 0.13
Fixed effects	Sex (male) × Density = 8	-0.04	-0.3, 0.21
Fixed effects	Diet (poor) × Density = 4	-0.03	-0.32, 0.27
Fixed effects	Diet (poor) × Density = 8	-0.09	-0.38, 0.21
Fixed effects	logP × Sex (male) × Diet (poor)	-0.68	-1.25, -0.11
Fixed effects	logP × Sex (male) × Density = 4	0.38	-0.11, 0.86
Fixed effects	logP × Sex (male) × Density = 8	0.24	-0.19, 0.66
Fixed effects	logP × Diet (poor) × Density = 4	0.04	-0.5, 0.58
Fixed effects	logP × Diet (poor) × Density = 8	0.17	-0.38, 0.73
Fixed effects	Sex (male) × Diet (poor) × Density = 4	-0.06	-0.43, 0.33

Section	Parameter	Estimate	95% CrI
Fixed effects	Sex (male) × Diet (poor) × Density = 8	-0.13	-0.51, 0.24
Fixed effects	logP × Sex (male) × Diet (poor) × Density = 4	0.11	-0.62, 0.83
Fixed effects	logP × Sex (male) × Diet (poor) × Density = 8	0.18	-0.56, 0.92
Among-mother SD	SD (Intercept)	0.03	0, 0.07
Among-mother SD	sd_id_mere__logP	0.05	0, 0.13
Among-mother correlation (r)	cor_id_mere__Intercept__logP	-0.21	-0.91, 0.76
Residual SD	Residual SD	0.12	0.12, 0.12

Table S2. Posterior summaries from the Bayesian morph-specific male allometry model. The model estimated Gaussian regression of log forceps length including body size (logPc), morph group, diet, density, and their interactions, with varying intercepts and slopes for mother identity (id_mere). Reported values are posterior medians and 95% credible intervals (CrI).

Section	Parameter	Estimate	95% CrI
Fixed effects	Intercept	1.21	1.08, 1.34
Fixed effects	logP	0.36	0.14, 0.59
Fixed effects	Morph (macrolabic)	0.26	0.07, 0.44
Fixed effects	Diet (poor)	0.07	-0.1, 0.23
Fixed effects	Density = 4	0.05	-0.14, 0.23
Fixed effects	Density = 8	0.03	-0.13, 0.19
Fixed effects	logP × Morph (macrolabic)	0.08	-0.24, 0.4
Fixed effects	logP × Diet (poor)	-0.17	-0.49, 0.16
Fixed effects	Morph (macrolabic) × Diet (poor)	-0.04	-0.33, 0.27
Fixed effects	logP × Density = 4	-0.09	-0.43, 0.24
Fixed effects	logP × Density = 8	-0.02	-0.31, 0.28
Fixed effects	Morph (macrolabic) × Density = 4	-0.01	-0.25, 0.25
Fixed effects	Morph (macrolabic) × Density = 8	0.08	-0.14, 0.3

Section	Parameter	Estimate	95% CrI
Fixed effects	Diet (poor) × Density = 4	-0.08	-0.31, 0.14
Fixed effects	Diet (poor) × Density = 8	-0.03	-0.24, 0.19
Fixed effects	logP × Morph (macrolabic) × Diet (poor)	0.08	-0.54, 0.68
Fixed effects	logP × Morph (macrolabic) × Density = 4	0.10	-0.34, 0.53
Fixed effects	logP × Morph (macrolabic) × Density = 8	-0.06	-0.45, 0.32
Fixed effects	logP × Diet (poor) × Density = 4	0.20	-0.24, 0.64
Fixed effects	logP × Diet (poor) × Density = 8	0.04	-0.41, 0.48
Fixed effects	Morph (macrolabic) × Diet (poor) × Density = 4	-0.41	-0.75, -0.06
Fixed effects	Morph (macrolabic) × Diet (poor) × Density = 8	-0.03	-0.4, 0.33
Fixed effects	logP × Morph (macrolabic) × Diet (poor) × Density = 4	0.52	-0.14, 1.2
Fixed effects	logP × Morph (macrolabic) × Diet (poor) × Density = 8	-0.15	-0.86, 0.57
Among-mother SD	SD (Intercept)	0.02	0, 0.09
Among-mother SD	sd_id_mere__logP	0.06	0.01, 0.19
Among-mother correlation (r)	cor_id_mere__Intercept__logP	-0.45	-0.99, 0.65
Residual SD	Residual SD	0.09	0.09, 0.09

56